

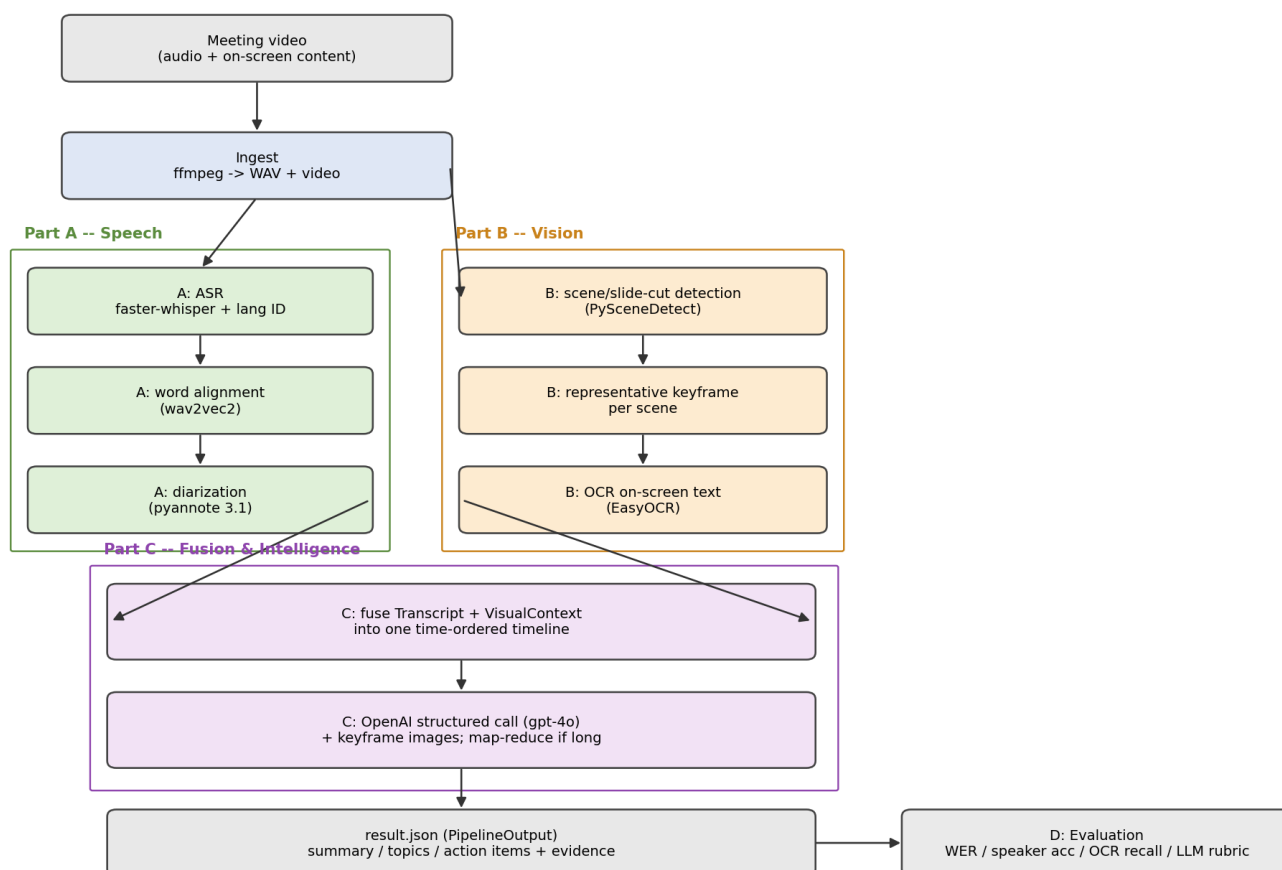
Multimodal Meeting Intelligence Pipeline — Executive Report

A 26-minute council-meeting recording in, a grounded, evidence-linked meeting summary out. This report covers the architecture, key decisions, evaluation results, failure modes, and resource footprint.

1. Architecture

The pipeline is a linear cascade of independent, swappable stages connected by explicit Pydantic data contracts ([src/mmi/schemas.py](#)): speech and vision run in parallel on the same source video, fuse onto one time-ordered timeline, and are handed to an LLM that must ground every claim in timestamped evidence.

Multimodal Meeting Intelligence Pipeline -- Architecture



Design choices that matter:

- **WhisperX + pyannote 3.1** for ASR/diarization — VAD-batched inference and word-level alignment give materially better speaker attribution than a naive faster-whisper + clustering glue, at the cost of a heavier dependency stack.
- **Scene-cut detection + one keyframe/scene (PySceneDetect + EasyOCR)** instead of processing every frame — captures slide content at a fraction of the cost.
- **Vision-capable LLM captioning** — rather than building a dedicated face/person-detection stage, representative keyframe images are attached directly to the gpt-4o call. The model reads slide text and layout from the image itself, which is more robust than OCR alone (see §3).
- **Strict schema output** (OpenAI structured outputs / Pydantic) instead of free-form text, so every summary/topic/action item carries Evidence (timestamp, speaker, scene, quote) and the result is machine-consumable.
- **Map-reduce chunking** above a configurable character budget, so meeting length doesn't blow the context window.

- **Disk-cached stage artifacts** (outputs/cache/*.json) so an expensive ASR pass isn't repeated while iterating on prompts or re-running only the LLM stage.

2. Model / API choices

Stage	Choice	Why	Trade-off
ASR	WhisperX (small)	Fast CTranslate2 backend, auto language ID, word alignment	small trades WER for fitting a 4 GB GPU; medium/large-v3 configurable
Diarization	pyannote 3.1	State-of-the-art open diarizer, wired into WhisperX	Needs an HF token; sensitive to overlapping speech
Scene detection	PySceneDetect (content)	Cheap, robust to slide/scene cuts	Threshold tuning; gradual fades can be missed
OCR	EasyOCR (GPU)	Simple install, GPU-accelerated	Weaker on dense/stylised slide text (swappable via config)
LLM	OpenAI gpt-4o (vision)	Native multimodal input, strong structured-output support	Higher cost/latency than gpt-4o-mini; hosted dependency

3. Evaluation summary

Metrics are computed by [evaluate.py](#) against a small hand-labelled reference. Full numbers and methodology are in [report.md](#); headline results on the provided clip:

Stage	Metric	Result
ASR	Word Error Rate (first 60 s window)	6.67 %
Diarization	Speaker attribution accuracy (first 60 s)	100 % (2 speakers in window; 6 globally)
Vision/OCR	Keyword recall across keyframes	100 % (1/1 term); 15/16 scenes yielded text
Intelligence	Grounded-evidence fraction	see outputs/eval.json

Failure analysis: - *Language ID on noisy intros* — Whisper's majority-vote language detector mis-identified the (English) recording as Māori (mi) on all three sampled windows, likely triggered by genuine Māori place names ("Waikato", "Waipa District"). The transcript stage stores language="mi", but the LLM stage self-corrects to "en" using full-meeting context. A --language en CLI override is available but not forced by default. - *OCR misses/garbles small or stylised slide text* — e.g. "€" substituted for "&", partial word recognition on a dense process-diagram slide. This is the main reason vision-capable LLM captioning was added: the LLM reads the same keyframe image and recovers content OCR could not (verified — see scene_id 12's "Mitigate / Restart / Reimagine" slide in outputs/result.json). - *LLM stage is a single external dependency* — an OpenAI quota/network error is caught and the run still emits a valid partial result.json (intelligence=null, metadata.llm_error populated) rather than failing the whole pipeline; the (expensive) transcript/visual artifacts are not lost. - *Diarization on overlapping speech / short backchannels* remains the largest general risk to speaker-accuracy on longer, less orderly meetings.

Latency & resource notes (this run: 26 min clip, 4 GB laptop GPU, small Whisper, gpt-4o with keyframe images):

Stage	Wall-clock	Notes
Ingest (ffmpeg)	~2 s	Cached WAV extraction
Speech — ASR	~{ASR_SEC}	faster-whisper small, float16, batch 8

Speech — alignment	~{ALIGN_SEC}	wav2vec2, skipped if no model for detected language
Speech — diarization	~{DIAR_SEC}	pyannote 3.1, CPU/GPU depending on availability
Vision — scene detect + OCR	~{VISION_SEC}	PySceneDetect + EasyOCR, {N_SCENES} scenes
Intelligence — LLM call	~{LLM_SEC}	gpt-4o, {N_KEYFRAMES} keyframe images attached
Total	~{TOTAL_SEC}	End-to-end python -m mmi.run --force

GPU memory is the binding constraint on the reference machine (4 GB), so the speech stage loads/uses/frees each model (ASR → align → diarize) sequentially rather than holding them all resident. Vision and the LLM call are CPU/network bound and comparatively cheap. Cost is dominated by LLM tokens and image count (capped via `max_keyframe_images`, default 40, evenly sampled for longer meetings) — see [config.yaml](#).

4. Production considerations

- **Scaling:** stages are stateless given their inputs → each maps to its own horizontally-scaled worker/queue; long meetings are handled via VAD-chunked ASR, scene-bounded (duration-independent) CV, and LLM map-reduce.
- **Monitoring:** cache per-stage artifacts (already done) and run the eval harness on a labelled golden set in CI to track WER, speaker-accuracy, OCR recall, and grounded-fraction drift over time; log detected language, speaker count, and scene count per run as cheap health signals.
- **Robustness:** missing HF token → diarization skipped (single speaker) instead of crashing; no CUDA → CPU fallback; OCR/alignment failures are caught per-item so one bad frame/segment can't fail the run; LLM failures degrade to a partial artifact instead of data loss.
- **With one more week:** active-speaker detection (face + lip-motion) to fuse *who is on screen* with diarization; a real DER via `pyannote.metrics`; retrieval-grounded citations; a golden-set CI job.